



An automatic calibration algorithm for laser vision sensor in robotic autonomous welding system

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Abstract

Visual sensor plays an important part in intelligentized welding systems, and the calibration of the vision sensor is the indispensable part of visual systems. Aiming at the problem of the tedious calibration process, this paper describes an automatic calibration algorithm. First, the robot motion equation and the motion range constraint equation are proposed to ensure that the collected images of calibration grid and laser line meet the calibration requirements. Based on these two equations, the automatic collection procedure can be realized. Second, the simplified visual servoing method and the Extended Kalman filter were used to adjust the images and rectify system parameters, respectively, which will improve the stability of the calibration motion. Third, to reduce the impact of the complex welding environments, a robustness feature extraction algorithm based on local threshold is studied. And then, the laser plane and hand-eye matrix are fitted with optimization algorithms to ensure calibration accuracy. Finally, the simulation experiments prove the feasibility and stability of the proposed algorithm. And the actual calibration tests suggest that the algorithm can significantly improve calibration efficiency. Moreover, the experimental results of welding guidance and seam tracking confirm that the calibration precision has met the welding requirement.

Keywords Robotic autonomous welding · Laser vision system · Automatic calibration · Robot motion planning · Image processing

Introduction

With the development of modern welding manufacturing, visual sensing technology has played an essential role in intelligentized welding systems (Chen 2015). Visual sensor based initial welding point guidance (Fan et al. 2019; Liu et al. 2018), seam tracking (Li et al. 2017; Ding et al. 2016), welding pool monitoring (Kwon et al. 2020; Wang and Li 2014) and welding quality controlling (Zhang and Chen

2017; Liu et al. 2019) are becoming the key technologies to realize intelligent welding. And the calibration of the vision sensor is always the prerequisite of vision systems, which has been broadly studied in recent years.

In the calibration process of the passive vision system, Ye et al. (2013) and Xu et al. (2017) described fast calibration methods that established a linear formula for coordinate transformation, when the pose of robot is fixed. Xiong et al. (2020) presented a simple calibration method by using a grid-pattern calibration plate. Diao et al. (2017) proposed an easy-to-use camera calibration methodology based on the information of the calibration grid and nozzle.

As for the laser vision sensor, the calibration process becomes more complicated. There are three parts of parameters to be determined: camera calibration, hand-eye calibration, and laser plane calibration. Suyi and Guorong (2008) put forward a fast calibration algorithm to calibrate camera parameters and hand-eye relationship simultaneously. Aiming at the problem of solving transformation matrices, Zou et al. used semidefinite programming (2018) and deep reinforcement learning (2020) to calculate the transformation matrices. By using specific calibration blocks, Abu-Nabah

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et al. (2016) presented a simple calibration method, which can calculate the transformation matrices with only one calibration image. Wang et al. (2020) introduced an efficient method that can calibrate the three sets of parameters simultaneously through minimal on-site operations.

Despite some achievements have been made in laser visual sensor calibration, most of the data acquisition processes during calibration are still performed by workers. Generally, these processes are complex and require a certain level of expertise and proficiency.

Therefore, the automatic calibration of visual system has gradually become an urgent problem to be solved in intelligent welding. The difficulty lies in that it is hard to control the camera to collect ideal calibration pictures through operating robot, when the hand-eye relationship is unknown. Especially for the visual sensors used for welding, they usually have a small field of view and high resolution to improve accuracy. And if the robot is moved randomly without any restraint, the calibration grid will probably move out of the view, which will lead to the failure of calibration.

To address the problem of tedious calibration process and low degree of automation, some automatic calibration methods were studied. Zhang et al. (2013) designed a camera calibration motion which can be conducted automatically by running the numerical controlled program. Wang et al. (2015) described an automatic procedure to calibrate a binocular system. Sharifzadeh et al. (2020) carefully defined a semi-automatic operation process for the hand-eye calibration of laser sensor. ANtonello et al. (2017) proposed a two-step calibration algorithm to determine the hand-eye relationship automatically based on a target detection algorithm. But it only focuses on hand-eye calibration and doesn't consider the loss of calibration objects, which will easily occur in small visual field cameras. Hu and Chang (2012) realized the automatic calibration of an eye-to-hand system based on structured light model. But the method is not suitable for the eye-in-hand system and is difficult to be used in the calibration process of laser vision sensor.

In this paper, aiming at the tedious and less automated calibration process of laser vision sensor, an automatic calibration algorithm is proposed. The main contributions include two parts: Firstly, two equations are proposed for automatic calibration motion planning, and a simplified image Jacobian and the Extended Kalman filter (EKF) are used to ensure the stability of the calibration process. Secondly, an adaptive image processing algorithm is developed to extract calibration data, and the laser plane and hand-eye matrix are solved by RANSAC algorithm and genetic algorithm (GA) respectively so as to ensure calibration accuracy. The details of the automatic calibration procedure will be explained in the later sections.

The remainder of this paper is organized as follows: section "[Vision system for robotic intelligent welding](#)"

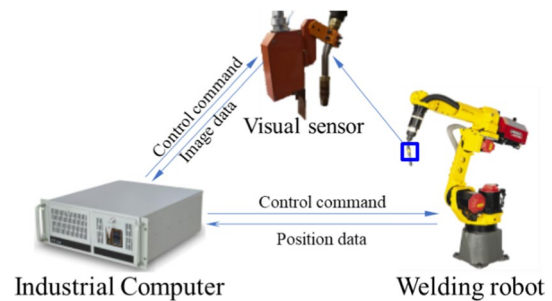


Fig. 1 Schematic diagram of robot intelligent welding vision system

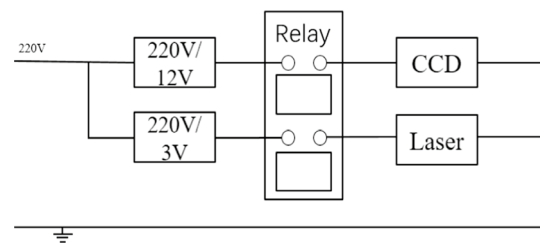


Fig. 2 Schematic diagram of the power module for laser vision sensor

introduces the developed laser vision system for robotic intelligent welding. In section "[Automatic calibration procedure](#)", the procedure of automatic calibration motion is discussed in detail. Then, the algorithms used for the calibration data processing are designed to minimize calibration error in section "[Transformation matrix computation](#)". Several experiments are conducted to test the performance of the proposed method in section "[Experimental verification and analysis](#)". Finally, the conclusions of this paper are summarized in section "[Conclusions](#)".

Vision system for robotic intelligent welding

Generally, robotic intelligent welding vision system mainly includes three parts: a laser vision sensor, an industrial personal computer (IPC), and a welding robot, as shown in Fig. 1. The vision sensor is rigidly mounted on the robotic manipulator, forming an eye-in-hand system. The sensor captures images and sends them to the IPC. A control program running on the computer will extract image features, and feed them back to the robot to form a closed-loop control.

The laser vision sensor mainly consists of an industrial camera, a structured line laser and a power module. A narrow band optical filter is installed to reduce arc interference during welding. To satisfy the demands of data acquisition in the calibration process, a relay is added to

control the power supply of camera and laser, as shown in Fig. 2. During each collection, the laser power supply is opened and closed once by relay, and the exposure parameter is modified simultaneously to collect one laser picture and one calibration grid picture, as Fig. 3.

In order to facilitate the subsequent description, each coordinate system is defined as follows: The robot base coordinate system is F_b , which is the world coordinate system simultaneously. The tool coordinate system is F_t , whose origin P_t is at the end of the torch. The camera coordinate system is F_c , whose origin P_c is at the center of the optical axis. The calibration grid coordinate system is F_m , and its origin P_m locates in the center of the calibration grid, as illustrated in Fig. 4.

As stated before, an accurate measurement model is a prerequisite of laser vision sensor based welding guidance and seam tracking. Given a point $m_l = [u, v, 1]^T$ in 2D image coordinate system F_l , its 3D coordinate $M_b = [X, Y, Z, 1]^T$ in F_b can be computed as:

$$M_b = \rho_b^t H_t^l H m_l \quad (1)$$

where ${}_b^l H$ presents the homogeneous transformation from F_b to F_l ; ${}_t^l H$ is the transformation matrix from F_l to F_t , which consists of two parts: ${}_t^l H = {}_t^c H * {}_c^l H$. And ${}_t^c H$ is the hand-eye matrix, which defines the transformation from F_t to F_c ; ${}_c^l H$ denotes the conversion matrix that maps a 2D laser point in frame F_l to its 3D coordinates in F_c .

In this paper, the conversion matrix ${}_c^l H$ is established with reference to the algorithm proposed by Huynh et al. (1999). Let the 3D laser coordinate system be F_l , and its origin is chosen in the centroid M of the calibration points set M_i^c . And the normal vector of the laser plane is regarded as the Z-axis of F_l . Based on Rodrigues' rotation formula, the rotation matrix ${}_c^l R$ of the F_{l13} relative to F_c can be found. In F_l , the Z coordinates of the points on laser plane are 0. So, a matrix S is adopted to eliminate the third component, to get a 2D coordinate system F_{l12} . Finally, with the homography H_k between F_{l12} and F_l , the conversion matrix ${}_c^l H$ from F_l to F_c can be obtained.

Camera calibration is the process of estimating camera parameters, including the internal parameters K and the

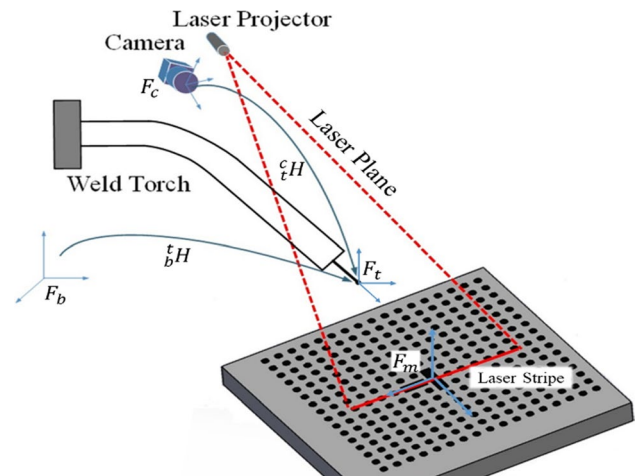


Fig. 4 Frames and transformation between them in system

extrinsic parameter ${}_c^m H$. K contains parameters of the camera's physical characteristics, such as focal length and principal point. And ${}_c^m H$ describes the location of the camera in F_m .

All parameters are listed in Table 1. It is evident that Eq. 1 doesn't contain the camera parameters. Since ${}_c^m H$ is the data basis for hand-eye calibration and laser plane calibration, camera calibration is still necessary.

Automatic calibration procedure

Calibration motion is a process of constantly changing the robot pose and driving the camera mounted on it to collect several sets of calibration data. During the calibration process, it is necessary to ensure that the captured pictures of calibration grid contain sufficient feature patterns to

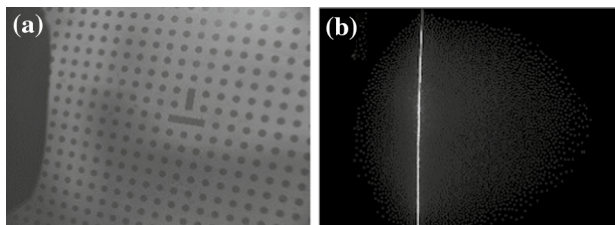


Fig. 3 The acquired images of calibration grid and laser

Table 1 Parameters of visual system models

Models	Parameters
Camera	$K = \begin{bmatrix} \alpha_x & 0 & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}, {}_c^m H = \begin{bmatrix} {}_c^m R & {}_c^m T \\ 0 & 1 \end{bmatrix}$
Hand-eye matrix	${}_t^c H = \begin{bmatrix} {}_t^c R & {}_t^c T \\ 0 & 1 \end{bmatrix}$
Structured light model	${}_c^l H = {}_c^l H^{-1} S^T H_k$ ${}_c^l H = \begin{bmatrix} {}_c^l R & \bar{M} \\ 0_{3 \times 1} & 1 \end{bmatrix}$ $S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

estimate camera parameters. And hand-eye calibration requires that the two relative motions of the robot are linearly independent, and the rotation axes are non-parallel. As for laser plane calibration, the laser plane is needed to keep intersecting with the calibration grid, and the poses relative to calibration grid are different.

By analyzing calibration requirements above, an automatic calibration procedure is designed as Fig. 5. It mainly includes three steps: Firstly, the exploratory movements are used to adjust the image of calibration grid and estimate system parameters. Secondly, two motion constraint equations are established based on the calibration needs, and the robot motion is planned to collect calibration data automatically. Finally, the transformation matrices will be fitted based on the collected data. This section will talk about the data acquisition procedure. And the calibration data processing will be discussed in section "Transformation matrix computation".

Robot motion equation

As mentioned above, it is difficult to plan the robot movement directly through camera motion due to the unknown hand-eye relationship. By analyzing the calibration requests, it is found that the robot needs to be moved randomly during hand-eye calibration. While the camera is fixed relative to the robot, the camera pose will change accordingly, which will meet the requirements of laser calibration. In this situation, the calibration task can be accomplished if the images of calibration grid collected during the movement meet camera calibration demands.

In this paper, a calibration grid with two rectangle feature patterns is used to establish the calibration grid coordinate system F_m . Thus, if the origin of F_m , marked as P_m , lies in the image center, the image of calibration grid can be guaranteed to contain abundant features to estimate camera parameters. Hence, this section proposes a robot motion equation by constraining the projection of P_m to achieve automatic calibration.

During the calibration process, the projection of a 3D point P in F_m , marked as p_i , in the i -th camera pose is formulated as Eq. 2:

$$p_i = sK({}^M R_i P + {}^M T_i) \quad (2)$$

If keeping ${}^M R_i P + {}^M T_i$ unchanged, p_i will remain a constant. And according to Fig. 4, the following transformation relationship exists among the coordinate systems:

$${}_b^M H = {}_b^t H * {}_t^c H * {}_c^M H \quad (3)$$

So, ${}_c^M T_i = {}_t^c R^{-1}({}_b^t R_i^{-1}({}_b^M T - {}_b^t T_i) - {}_t^c T)$ and ${}_c^M R_i = {}_t^c R^{-1}{}_b^t R_i^{-1}{}_b^M R$. Thus, Eq. 4 can be derived.

$${}_b^t R_i^{-1}({}_b^M R P + {}_b^M T - {}_b^t T_i) = const \quad (4)$$

Since the coordinate of P_m is $[0001]^T$ in F_m , Eq. 4 can be simplified as Eq. 5, which is the robot motion equation:

$${}_b^t R_i^{-1}({}_b^M T - {}_b^t T_i) = const \quad (5)$$

It is worth noticing that Eq. 2 doesn't specify the image coordinate of p_i . It means that p_i only needs to be maintained in the middle of image to ensure that the calibration image has sufficient features.

In Eq. 5, ${}_b^M T$ is fixed, and the variables are only related to the pose of robot. So, this equation can be used to plan robot motion directly: When ${}_b^t R_i$ is given, the corresponding ${}_b^t T_i$ will be obtained. Even in the case of unknown hand-eye relationship, Eq. 5 can ensure that the images collected during the planned robot movement will satisfy the needs of calibration.

Additionally, by comparing Eqs. 4 and 5, how the error of ${}_b^M T$ affects Eq. 5 can be discovered: If the measurement of ${}_b^M T$, denoted ${}_b^M \hat{T}$, has an error $dT = {}_b^M \hat{T} - {}_b^M T$, it's no longer to keep the projection of P_m unchanged, but to keep the projection of a point near P_m invariable. And the point's coordinate is $P = {}_b^M R^{-1}dT$. When the error is small, Eq. 5 can still ensure that the captured images will meet the calibration requests.

Restriction of robot motion range

The equation proposed above only guarantees the acquisition of calibration grid images. However, it doesn't take into account the change of the laser plane caused by robot motion. If the robot moves without restraint, the intersection

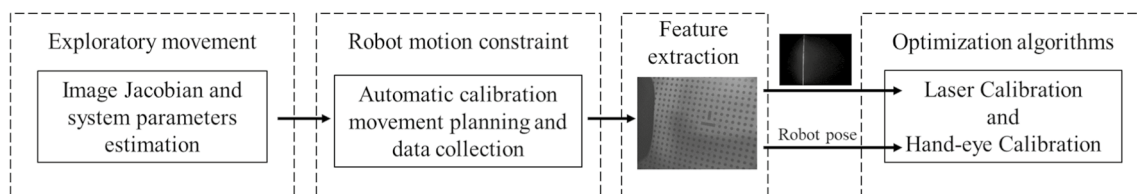


Fig. 5 The flowchart of the auto-calibration procedure

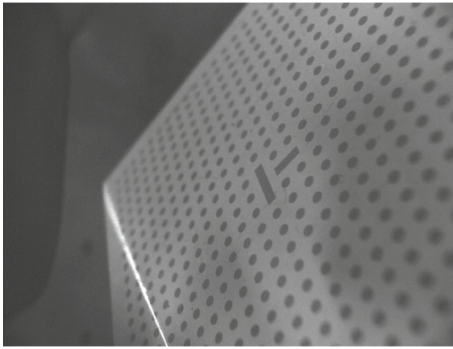


Fig. 6 Calibration images acquired in oversized camera pose

line between the laser plane and the calibration grid, i.e. laser line, may be far away from the center of calibration grid, or even move out of view, as expressed in Fig. 6. For this reason, it's needed to add another constraint equation to limit the range of robot motion.

As the laser plane is fixed relative to the camera, the projection of laser line mainly depends on its distance from the camera. And the distance can be reflected from the pose of camera relative to the calibration grid, i.e. ${}^cM H$. When the pose changes greatly, it is possible for the laser line to move out of view. Furthermore, the angle between the Z-axes of F_c and F_M has much greater influence on the distance. Therefore, by limiting the angle, the problem that laser line is too short owing to oversized camera poses can be prevented.

Let the rotational part of calibration grid pose be cMR_i in the i -th position. According to the geometric interpretation of rotation matrix, the Z-axis of F_M is $Z_m = {}^cMR_i * Z_c$, where Z_c stands for the Z-axis of F_c . So, the motion range constraint equation can be formulated as Eq. 6:

$$\theta = \cos^{-1}(Z_m, -Z_c) < \theta_{max} \quad (6)$$

where minus is to ensure that the result is an acute angle, and θ_{max} is the maximum allowable angle.

However, before arriving at the i -th position, cMR_i is unknown. Hence, it's essential to establish the relationship between cMR_i and robot motion. Based on Eq. 3, there is:

$${}^cMR_i = {}^cMR_{i-1} * {}^bMR' * \Delta_b^tR * {}^bMR \quad (7)$$

where $\Delta_b^tR = {}^bR_{i-1} * {}^bR'_i$ can be calculated from the planned robot motion, and bMR is estimated by exploratory motion, which will be discussed later.

It is worth mentioning that Eq. 6 is only used to ensure the laser line won't move out of the calibration grid and isn't directly involved in the motion planning. Therefore, the estimation error of bMR doesn't affect the stability of the calibration motion to some extent. If the error results in the estimated angle $\hat{\theta}_i > \theta_i$, the allowable range of camera motion will be reduced, but the calibration motion within the feasible range

won't be affected. And if the error makes $\hat{\theta}_i < \theta_{max} < \theta_i$, the allowable range of motion will increase. Since θ_{max} is a manually preset threshold, it is possible to make the limited range slightly smaller than the actual allowable range by appropriately decreasing the threshold. So, the collected picture can still meet the calibration requirements when the angle exceeds the restricted range. For this reason, the constraint equation doesn't require an accurate estimation of bMR .

Parameter estimation and rectification

According to the analysis described in sections "Robot motion equation" and "Restriction of robot motion range", it is found that the pose of F_c in the world coordinate system F_m is the basis of motion control. Thus, this section will introduce exploratory motions to estimate bMH .

Firstly, the robot is arbitrarily moved twice within a small range of initial position. For the relative motion of robot and camera, there exists the following relationship:

$${}^bMH\Delta_c^MH = \Delta_b^tH {}^bMH \quad (8)$$

where Δ_c^MH and Δ_b^tH are the relative motion of camera and robot, respectively.

According to Navy algorithm (Park and Martin 1994), bMH can be solved. Since the data set is relatively small and Δ_c^MH may exist estimation error, bMH is only a rough guess and needs further rectification.

Since bMR is invariable, the Euler angle associated with bMR , noted as $V = (\alpha, \beta, \gamma)'$, will also stay unchanged during calibration. According to Eq. 7, the state equation and observation equation of the bMR are established:

$$\begin{cases} V_i = V_{i-1} + p \\ \theta_i = h(V_i, \Delta_b^tR_i) + q \end{cases} \quad (9)$$

where $h(V_i, \Delta_b^tR_i) = \arccos(-[001] * {}^cMR_1 * {}^bMR' * \Delta_b^tR_i * {}^bMR * [001]')$; And p and q are the process and measurement noise, respectively, whose covariance are R and Q , respectively.

Given that Eq. 9 represents a nonlinear system, the EKF algorithm is applied to rectify bMR during calibration motion. The algorithm adopts the first order Taylor series expansion (Jacobian matrix) to approximate a nonlinear function linearly.

According to EKF algorithm, when the system is in the i -th position, the prior estimate of V_{i+1} is $\hat{V}'_{i+1} = V_i$. So, the prediction of V_{i+1} is derived as follows:

$$\hat{V}_{i+1} = \hat{V}'_{i+1} + K_{i+1}(\theta_{i+1} - H\hat{V}'_{i+1}) \quad (10)$$

where H is the Jacobian matrix of h , and K_{i+1} is the Kalman gain:

$$K_{i+1} = \frac{\hat{p}_{i+1} H^T}{(H \hat{p}_{i+1} H^T + R)} \quad (11)$$

As discussed previously, the motion constraint equations have certain robustness. And with the help of EKF algorithm, the estimated parameters will be corrected during calibration movement. So, the estimation error won't affect the stability of calibration process to some extent.

Calibration image adjustment

Because of the estimation error of ${}^b_M T$, the projection of the origin of calibration grid, marked as p_m , will change in different robot positions. So, an image adjustment motion is planned to adjust p_m , to keep the calibration motion stable.

Let the coordinate of image center be p_0 . So, the expected image motion is $p_0 - p_m$. In the adjustment motion, only the robot motions in X and Y directions are considered, marked as dx and dy , respectively. And the image motion and robot motion are related by the image Jacobian matrix, noted as J , as expressed in Eq. 12.

$$p_0 - p_m = J * \begin{pmatrix} dx \\ dy \end{pmatrix} \quad (12)$$

The image Jacobian is estimated by exploratory movement. The robot is controlled to move along X and Y directions. And the corresponding motions of p_m are recorded as p_x and p_y , respectively. So, the image Jacobian in the initial pose is $J = [p_x \ p_y]$. As J is invertible, the robot motion required to adjust p_m to the image center can be obtained by solving Eq. 12, as Fig. 7.

In Eq. 12, the Jacobian matrix is a simplified form that only considers the velocity in X and Y directions. And it depends on the hand-eye matrix and the depth of the feature point (Gong et al. 2017). As the depth changes slightly and

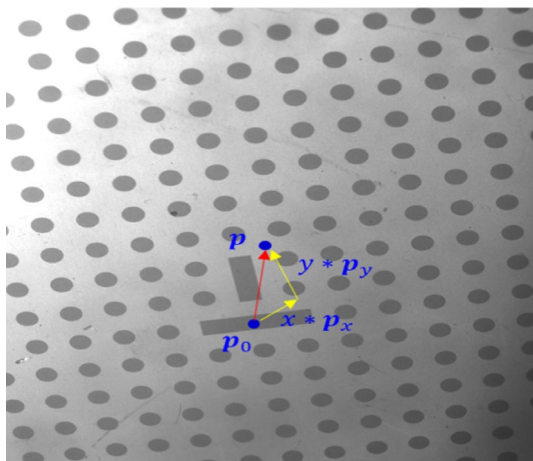


Fig. 7 Schematic diagram of image adjustment

${}^t_c H$ remains unchanged during the robot motion, the matrix can be treated as a constant in different robot poses. So, the image Jacobian can be used to adjust p_m to image center after each robot motion.

The procedure of automatic data collection

At the end of this section, based on the conclusions drawn above, the recommended flow of automatic calibration procedure is described as follows:

1. Move the camera above the calibration grid, and estimate the ${}^b_M R$, ${}^b_M T$ and J by exploratory motions.
2. Select the robot rotation motion ${}^b_t R$ arbitrarily and calculate the corresponding translational part according to Eq. 5. And check whether the robot motion range constraint is satisfied.
3. Control the robot to ${}^b_t H$, and adjust the image according to Eq. 13 if necessary.
4. Collect calibration data and rectify ${}^b_M R$ with EKF algorithm.
5. Repeat steps 2–4 until enough calibration data is collected.

To meet the requirement of hand-eye calibration, which is the two relative rotations of the robot are not coaxial, the rotation motion ${}^b_t R$ of the robot is planned as:

1. Randomly rotate the robot $0\text{--}3^\circ$ about the Z -axis of the world coordinate system F_b ;
2. Randomly rotate the robot around the X and Y axes of F_b within $\pm 5^\circ$.

Transformation matrix computation

Another important step to realize automatic calibration is to calculate the laser plane equation and the hand-eye matrix automatically. It mainly includes two parts: image feature extraction and transformation matrices computation. The accuracy of the extraction and computation results directly determine the accuracy of calibration. This section will propose a robust image processing algorithm to extract features and discuss the optimization algorithms used for conversion matrices fitting.

Calibration image feature extraction

As an essential part of calibration data, the images of calibration grid are used to determine the pose of camera. Therefore, the success of feature extraction is directly related to the success of calibration.

For laser vision sensors used in welding environments, the narrow band optical filter will significantly reduce the light received by the camera. Even by increasing the exposure time, the captured image will still be dark without using additional light sources. Meanwhile, there will exist a small amount of welding noise on the protection glass, such as welding spatter and dust, which will seriously affect the calibration images. In addition, the uneven illumination in the actual environment may give rise to local shadow, which will also lead to a breakdown of the image quality.

It is apparent that the lighting condition is the critical factor affecting image quality. To overcome the impact of varying illumination on images, the Sauvola local threshold algorithm (Sauvola and Pietikäinen, 2000) is used to extract feature patterns. The mean m and variance s of the gray value are computed in a neighborhood of each pixel. And the local threshold T is computed according to Eq. 13, where k denotes a correction coefficient. If the grayscale of a pixel is smaller than T , it is considered to be the target, as Fig. 8b.

$$T = m + \left(1 + k * \left(\frac{s}{125} - 1\right)\right) \quad (13)$$

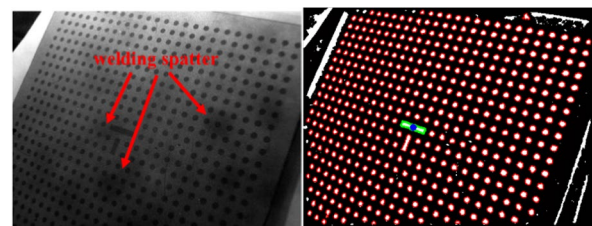
Considering that the calibration grid may be untidy and the image may have noise, the binarized image will exist some interference, which can be treated as pepper-salt noise. So, the median filter is used to remove the disturbance. And then, the opening operator is adopted to cut off some short connections between adjacent areas to prevent their impact on identifying the feature patterns, as expressed in the upper left of Fig. 8b. Through the processing above, a pure binary image of calibration grid is obtained, and the feature patterns are displayed in white in Fig. 8c.

Finally, every connected component is extracted and recorded via traversing the image. Then, all components are sorted according to their area, and only 25% to 80% part of the queue, i.e. the circular patterns, are left, presented as the areas surrounded by red edges in Fig. 8c. And the centroid of circular pattern is the desired feature point for conversion matrices fitting. In the calibration pattern used, the length

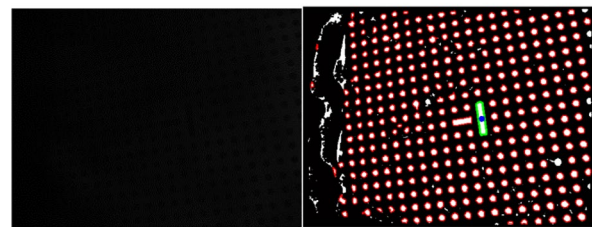
of the two rectangles is 3 and 5 times the circular diameter, respectively. Therefore, the largest rectangle is the region with an area between 4.5 and 7.5 s , where s is the average area of the circular patterns, as shown the rectangle with green edges in Fig. 8c. And its centroid is regarded as the origin of F_m , as the blue dot.

For the welding spatter stuck to the protection glass, its influence on the image can be also regarded as uneven lighting. So the extraction algorithm will still be effective. As shown in Fig. 9a, the algorithm can deal with local black spots caused by spatter. With regard to the images with low intensity caused by the filter, insufficient ambient light, or short exposure time, the algorithm still maintains a good effect, as shown in Fig. 9b. Since the local threshold of a pixel depends on the grayscale distribution in its neighborhood, the algorithm has certain adaptability to the inadequate lighting condition.

To sum up, the proposed feature extraction algorithm has good robustness and can cope with the complex



(a) The extraction result of image with welding spatter



(b) The extraction result when the image is too dark

Fig. 9 The effect of feature extraction algorithm in complex environment

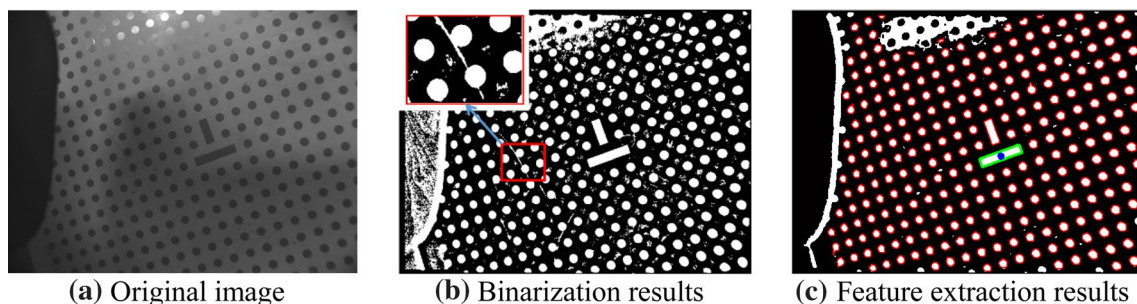


Fig. 8 The feature extraction process of calibration image

welding environment. It will provide an effective guarantee for the stability of automatic calibration process, and help to reduce manual operations, such as removing filter and providing additional light sources.

Solution for hand-eye matrix

The traditional solution algorithms for hand-eye matrix searched the rotational and translational parts separately and required the camera extrinsic parameters. Consequently, the estimation errors of camera extrinsic parameters, rotation matrix and translation matrix will accumulate gradually, and lead to a certain error in the results. Heller et al. (2015) introduced a branch-and-bound algorithm to solve the hand-eye matrix directly without the extrinsic parameters in pixel space. However, it searches for the optimal solution globally, which may be time-consuming. Therefore, this section puts forward a two-step method to solve the hand-eye matrix. Firstly, the traditional Tsai algorithm (Tsai and Lenz 1989) is used to estimate the hand-eye matrix. And then, the optimal solution is searched in the pixel space by GA.

According to the epipolar constraint, a pair of corresponding feature points u and v in two calibration pictures satisfies Eq. 14:

$$v^T (K^{-T} * \Delta_c^M T \times \Delta_c^M R * K) * u = 0 \quad (14)$$

where K is the internal matrix of camera; $\Delta_c^M R$ and $\Delta_c^M T$ are the relative rotation and translation of camera, which can be written as Eqs. 15 and 16, respectively:

$$\Delta_c^M R = {}^c R \Delta_b^t R {}^t R {}^c R^{-1} \quad (15)$$

$$\Delta_c^M T = {}^c R^{-1} ((\Delta_b^t R - E) * {}^t T + \Delta_b^t T) \quad (16)$$

Substituting Eqs. 15 and 16 into Eq. 14, the extrinsic parameters of camera are isolated. In this paper, the median value is taken as the target to avoid the interference caused by the feature extraction error. And the problem of solving hand-eye matrix is converted to solving the following objective function:

$${}^c R, {}^c T = \min \text{mid} \left(v_{ij}^T (K^{-T} * \Delta_c^M T_i \times \Delta_c^M R_i * K) * u_{ij} \right) \quad (17)$$

Table 2 Parameter settings

Parameters	Value
Crossover probability	0.6
Mutation probability	0.02
Population size	100
Max generations	500

where i and j represent the j -th pair corresponding points of the i -th relative movement, respectively.

Equation 17 is solved by the genetic algorithm, and the parameters are set as Table 2. The initial population of GA is set as the result of the Tsai's method, marked as ${}^t H_0$. And the optimal solution of Eq. 17 is searched within a specific range of ${}^t H_0$. This new two-step method is a further modification of Tsai algorithm in pixel space. As a result, the effect of the accumulated error will be reduced.

Laser plane fitting

Before fitting the laser plane, it is necessary to obtain the 3D coordinates of laser points. To begin with, the coordinates of feature points in the calibration grid frame F_M and image are both known, marked as G_i and g_i , respectively. Next, the equation of the laser line is extracted and fitted through image processing, shown as the blue line in Fig. 10. And the intersections of the laser line and every row (or column), noted as p , are the points used for laser plane fitting, as the purple dots in Fig. 10.

According to the invariance of the cross-ratio, the 3D coordinates of p in F_M , marked as P , can be obtained as Eq. 18:

$$\left(\frac{PG_{i-1}}{PG_i} \right) \left(\frac{G_{i+1}G_{i-1}}{G_{i+1}G_i} \right) = \left(\frac{pg_{i-1}}{pg_i} \right) \left(\frac{g_{i+1}g_{i-1}}{g_{i+1}g_i} \right) \quad (18)$$

A pair of calibration grid image and laser line image only provides one set of collinear laser points. However, laser plane fitting requires multiple groups of non-collinear laser points. To achieve this, the camera extrinsic parameters are used to convert the 3D points to camera frame F_c , as Eq. 19. All the extracted laser points are drawn in Fig. 11.

$$M_{ij}^c = {}^c H_i^{-1} P_{ij} \quad (19)$$

where ${}^c H_i$ stands for the camera extrinsic parameter corresponding to the i -th image of calibration grid; P_{ij}

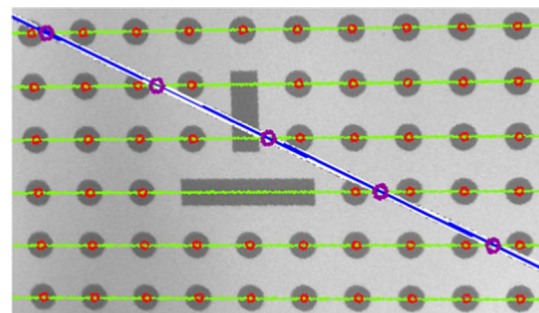


Fig. 10 The extracted 2D coordinates of laser points

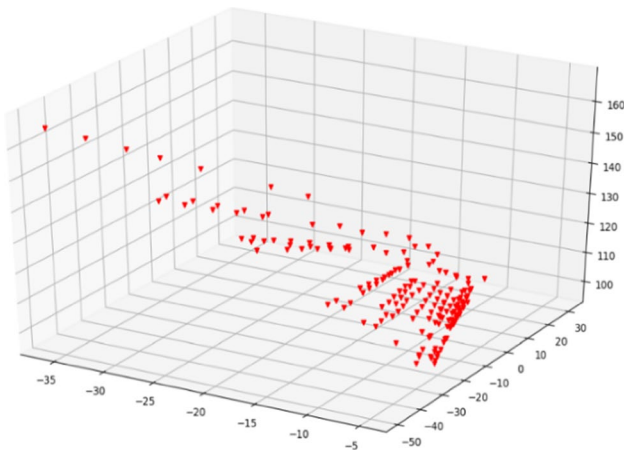


Fig. 11 The 3D laser points in camera frame

represents the j -th laser point in F_M at the i -th position; and M_{ij}^c is the j -th laser point in F_C in the i -th position.

During the planned calibration procedure, ${}^cM^T$ is kept unchanged, while cR is different in various positions. With F_C as the reference, the location of P_m remains unchanged, while the X - O - Y plane of F_m rotates around P_m . For the reason that the laser plane is fixed relative to the camera, the intersection line between the calibration grid and the laser plane must be different at various positions.

Because of the estimation error of ${}^cM^T$ and the extraction error of feature points, M_{ij}^c may contain noises and not be entirely on the same plane. Therefore, the RANSAC algorithm is used to fit the plane equation. The target function is $Ax + By + Cz + 1 = 0$. Three points in the point set are randomly selected to fit a plane equation. And if the distance between more than 90% of the points in the point set and the plane is less than a specified threshold, the equation is considered the laser plane equation.

Experimental verification and analysis

The experimental system

Finally, experiments are conducted to verify the proposed method. The experimental system used in this paper mainly includes an IPC, a laser vision sensor, a FANUC welding robot, and a Fronius CMT welding power supply, as shown in Fig. 12. The IPC connects the robot via Robot Interface so as to read robot's position, and write register and position variables in real time.

The interface of the developed program is shown in Fig. 13. The software can complete the automatic calibration task by cooperating with a simple program on robot.

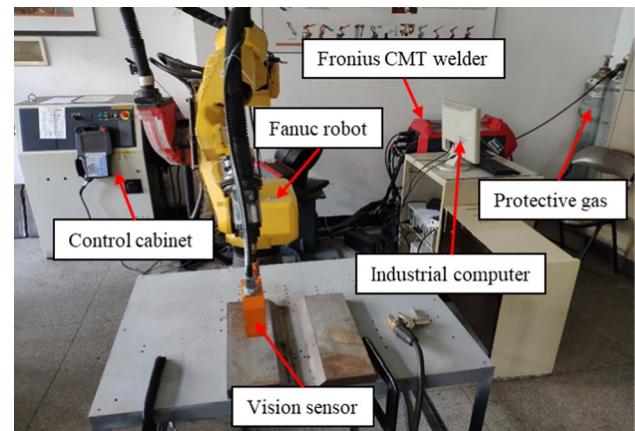


Fig. 12 Visual sensing experimental system

During the calibration process, the program sends the newly calculated position to the robot. When the robot arrives at the designated position, it sends an acquisition command to the program. The program collects the calibration data and sends a new position to robot again. After completing the acquisition motion, the transformation matrices will be calculated automatically.

Simulation of the calibration process

Simulation experiments of the calibration process are performed to investigate the feasibility and stability of the proposed method. In the experiment, it is assumed that the estimation errors of ${}^bM^T$ and bR are $dT = \|{}^b\hat{M}^T - {}^bM^T\|_2$ and $dR = \|\hat{V} - V\|_2$, respectively, where V stands for the Euler angle converted from bR .

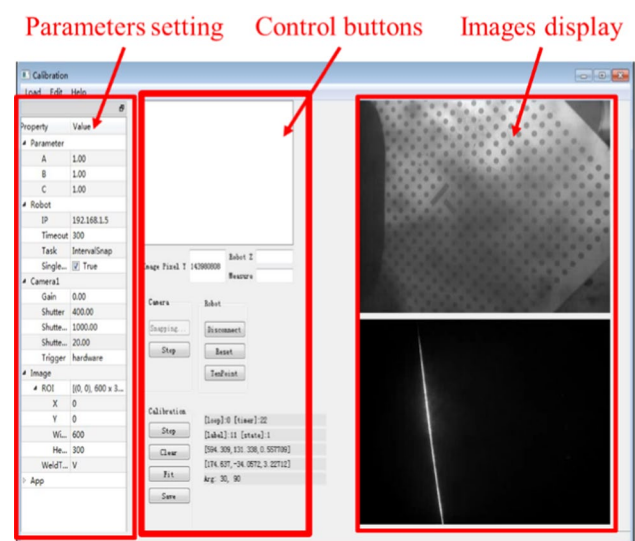


Fig. 13 Automatic calibration program

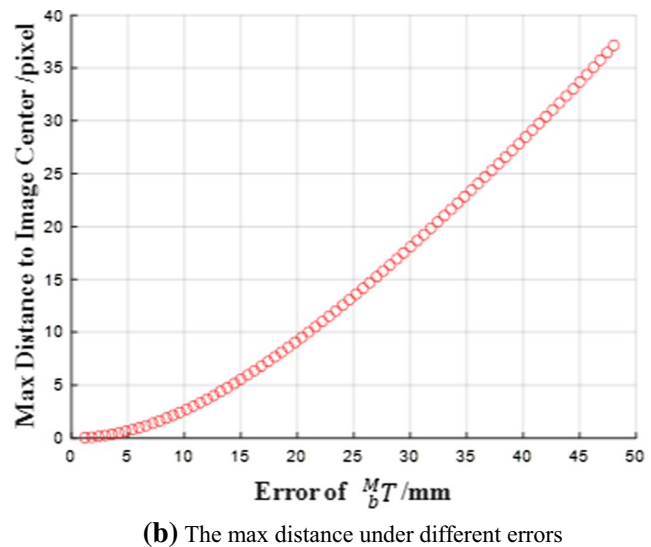
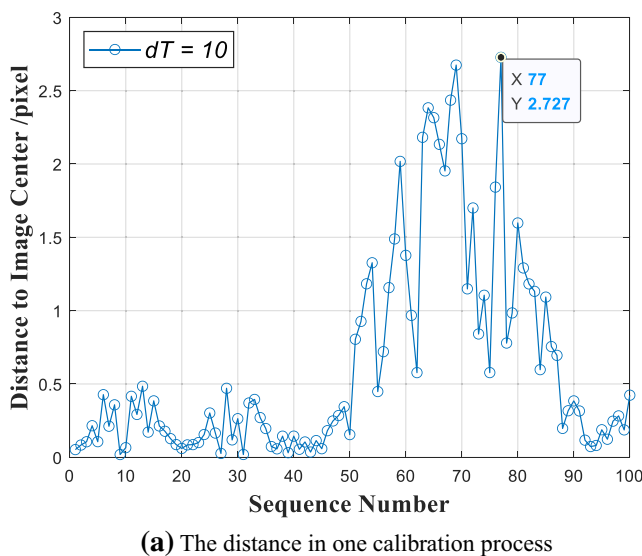


Fig. 14 The influence of dT on image acquisition

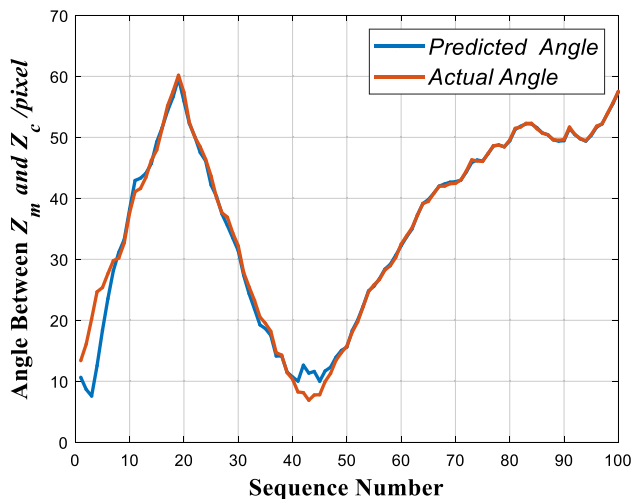


Fig. 15 The predicted angle and actual angle between Z_m and Z_c when $dR = 15^\circ$, $\theta_{max} = 60^\circ$

Firstly, the influence of dT on the image acquisition process was validated, and the results are shown in Fig. 14. The distance between the projection of P_m and the image center was used to evaluate the feasibility of collection process. As shown in Fig. 14a, in one calibration process with $dT = 10$ mm, the maximum distance is less than 3 pixels. Thanks to the image adjustment movement, the projection of P_m will be kept in the image center. So, all the acquired images can meet the calibration requirements. And the max distances under different errors are shown in Fig. 14b. It is found that the max distance will grow with the increase of dT . But even with the error of $dT = 48$ mm, the distance is less than 37 pixels. It turns out that

the calibration process has a good robustness against the estimation error of ${}_b^M T$.

And then, the influence of dR on motion range constraint equation (Eq. 6) is illustrated in Fig. 15. With the help of EKF algorithm, the predicted value will be close to the actual one after several iterations. The dR will only affect the initial value of EKF. And before reaching the limitation, the prediction error has become quite small. This means that the EKF algorithm can guarantee the stability of calibration process.

Generally, the estimated ${}_b^M \hat{T}$ and ${}_b^M \hat{R}$ are slightly different from their actual values. In actual experiments, it is found that dT is below 9 mm, and the maximum estimation error of V is less than 6° . The test results reveal that dT and dR have little influence on the stability of the calibration process to some extent. So, the estimated ${}_b^M H$ can be used in the motion equations directly.

Automatic calibration experiment and error analysis

With the auto-calibration procedure described in section "Automatic calibration procedure", the robot can automatically collect sufficient images that meet calibration demands in a short time. The camera poses during calibration motion are presented in Fig. 16b. The entire calibration process only needs to move the robot manually to the initial position, which greatly reduces manual intervention compared to the traditional method (Wang et al. 2020; Abu-Nabah et al. 2016).

By using the method presented in this paper, it can easily collect over 30 sets of data within 5 min. But using the method described by Wang et al. (2020), the calibration process gathered 10 sets of calibration images in 10 min

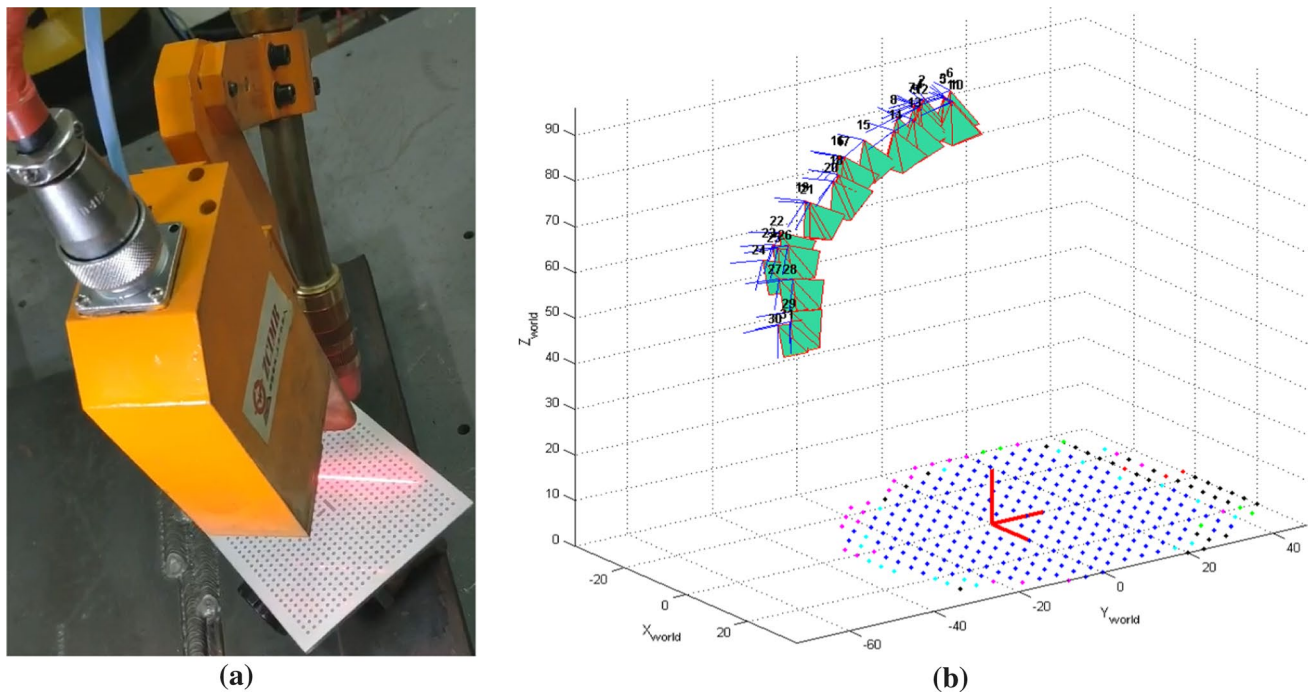
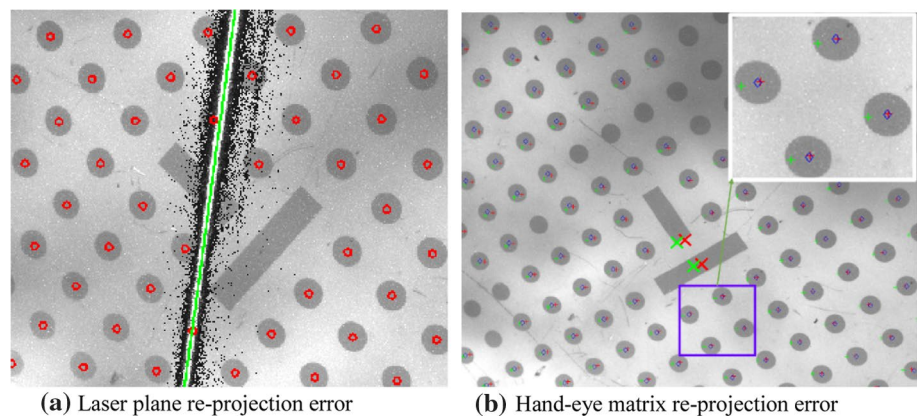


Fig. 16 **a** The Automatic calibration experiment. **b** The collected camera poses during experiment

Fig. 17 Re-projection error analysis



(a) Laser plane re-projection error

(b) Hand-eye matrix re-projection error

and need several manual operations. It is suggested that the efficiency of calibration process has been greatly improved.

And then, the re-projection experiments were carried out to measure the calibration error qualitatively. Firstly, the laser plane was re-projected into the image. As shown in the green line in Fig. 17a, the laser plane is basically in the center of the laser line. Afterwards, the points on the calibration grid were re-projected into the image with the hand-eye matrix, as Fig. 17b. The re-projection results of the traditional Thai algorithm and the algorithm used in this paper are marked in green and red, respectively. And the actual feature points are shown in purple. The result indicates that the two-step algorithm can improve the accuracy of the hand-eye matrix. Generally, the average

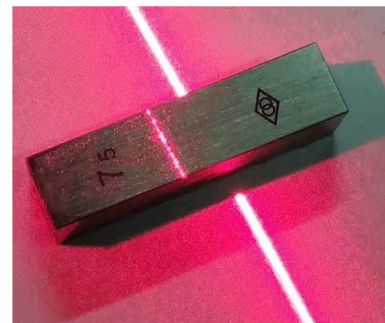


Fig. 18 Measure standard block to verify laser calibration accuracy

re-projection error is less than 2 pixels, and the corresponding actual distance is about 0.2 mm.

Furthermore, the measurement experiments were conducted to verify the laser calibration accuracy quantitatively. The length W and height H of a standard block were measured in camera frame F_C at different angles, as shown in Fig. 18. The actual W and H of the standard block are 7.500 mm and 9.000 mm, respectively. And the results are summarized in Table 3. The measurement error is between 0.03 and 0.06 mm.

Verification of calibration accuracy

To verify the calibration accuracy, the welding guidance experiment and seam tracking experiment were carried out with the calibrated sensor as shown in Fig. 19, and the guidance error and tracking error were used to determine the static precision and dynamic precision of the calibration result, respectively.

Welding guidance experiment

The calibration accuracy of the visual sensor was verified via the welding guidance test. By scanning the workpiece with the calibrated sensor, the point cloud data was reconstructed. The initial welding point is extracted through point cloud processing. And the actual position of the initial point is collected through robotic teaching. Ten sets of guidance tests were conducted in different camera postures. Test results indicate that the guidance error is between 0.37 and 0.57 mm, as shown in Fig. 20.

Seam tracking experiment

The seam tracking experiment was conducted to evaluate the dynamic accuracy, and the welding parameters is shown in Table 4. 200 sampling points were taken on a welding seam. The 3D coordinates of these points are acquired with the

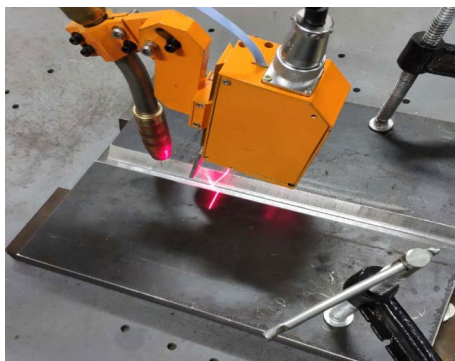


Fig. 19 Verifying the calibration accuracy in welding experiment

Table 3 Standard block measurement error

	Pose 1	Pose 2	Pose 3
Error of W (mm)	0.04	0.05	0.03
Error of H (mm)	0.03	0.06	0.05

calibrated laser vision sensor, as Eq. 1. Similarly, the actual coordinates were recorded through teaching. The Euclidean distances between the calculated values and the actual ones are treated as tracking error.

Moreover, to show the advantage of laser vision sensor based seam tracking, after recording the actual welding path, deviation was added to the teaching trajectory, as Line-A in Fig. 21a. During welding, the visual sensor, which has been calibrated by the proposed algorithm, was used to correct the welding trajectory (Line-B). The results show that compared with the traditional teaching-and-playback robot, the robot with the calibrated sensor can still complete the welding when the weld seam deviates from the teaching trajectory, as Fig. 21b. And the average tracking error is about 0.3 mm, as shown in Fig. 22.

Therefore, the calibration accuracy can satisfy welding guidance and seam tracking requirements and meet the needs of general welding production, which proves that the calibration method has certain practicability.

Conclusions

Aiming at the problem that the calibration process of laser vision sensor is complex and inefficient, an automatic calibration algorithm is presented in this paper. The following conclusions are drawn:

1. After analyzing the calibration requirements, the robot motion equation and the motion range constraint equation are derived for robot motion planning. Based on the

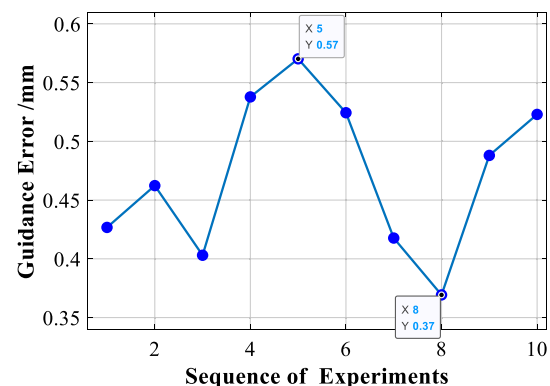
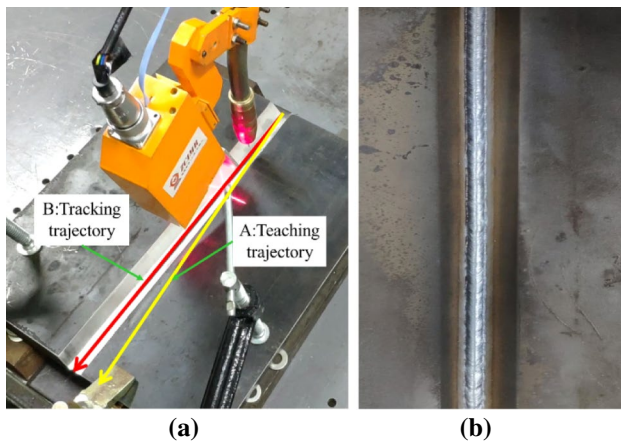
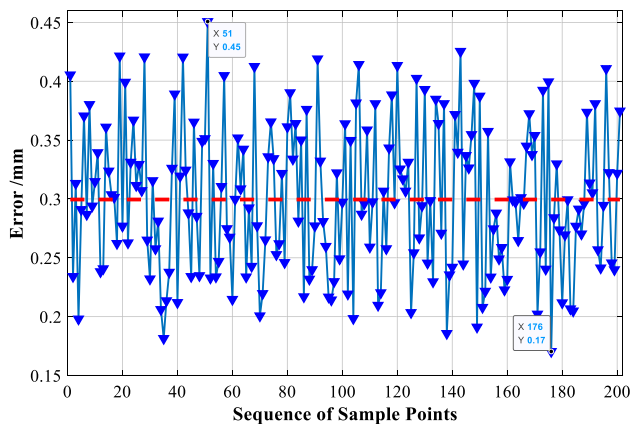


Fig. 20 The accuracy of welding guidance

Table 4 Welding parameters

Parameters	Value	Parameters	Value
Material	Q235 steel plate	Welding speed	4 mm/s
Pulse frequency	125 HZ	Welding wire	1.2 mm
Welding current	130 A	Protective gas	80%Ar + 20%CO ₂
Dry elongation	15 mm	Gas-flow rate	14L/min

**Fig. 21** a Seam tracking experiment. b The tracking result**Fig. 22** The accuracy of seam tracking

two equations, it's possible to realize automatic calibration data collection only through operating robot.

2. A simplified image Jacobian matrix is used to adjust the image position, and the Extended Kalman filter is adopted to rectify system parameters, so as to improve the stability of the calibration process.
3. To achieve desired calibration accuracy, an adaptive image processing algorithm based on local threshold algorithm is designed for feature extraction. The algorithm can maintain good robustness even under complex welding environments. Besides, the laser equation is fitted by RANSAC algorithm, and the hand-eye matrix is

calculated by genetic algorithm in pixel space to obtain satisfactory calibration accuracy.

4. The simulation results prove the stability and feasibility of the presented algorithm. In actual calibration test, the calibration error is about 0.2 mm. Moreover, the results of welding guidance and seam tracking experiments indicate that the static precision and dynamic precision of calibration result are sufficient to satisfy most welding demands.
5. The proposed automatic calibration algorithm can easily collect over 30 sets of data within 5 minutes, which suggests that it is a highly efficient method compared to the traditional calibration method. Meanwhile, the entire calibration process only needs one manual operation, which greatly reduces the requirement of proficiency for workers.

Additionally, the proposed algorithm can also be used to calibrate the passive vision sensor automatically. In the future, this method will be further simplified to promote its application in vision based intelligent manufacturing systems.

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